

Name: \_\_\_\_\_

Class: \_\_\_\_\_

# SYDNEY TECHNICAL HIGH SCHOOL

## YEAR 12

### HSC ASSESSMENT TASK 3

JUNE 2009

### MATHEMATICS Extension 1

**Time Allowed:** 70 minutes

**Instructions:**

- Write your name and class at the top of each page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a **new page**.
- Standard integrals can be found on the last page.

| Question 1 | Question 2 | Question 3 | Question 4 | Question 5 | Question 6 | Total |
|------------|------------|------------|------------|------------|------------|-------|
| /10        | /11        | /10        | /10        | /11        | /11        | /63   |

**Question 1****Marks**

- a) Solve  $\log_e x + \log_e(x - 3) = \log_e 4$  3
- b) Evaluate and leave in exact form
- (i)  $\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$  1
- (ii)  $\sin\left(2 \cos^{-1}\frac{4}{5}\right)$  2
- c) Find (i)  $\int \frac{2}{9+x^2} dx$  2
- (ii)  $\int \frac{dx}{\sqrt{1-9x^2}}$  2

**Question 2**

- a) Consider the function  $f(x) = 8x - x^2$
- (i) Sketch  $y = f(x)$  clearly showing the  $x$  and  $y$  intercepts and the vertex. 2
- (ii) State the largest domain containing  $x = 8$  for which  $f(x)$  has an inverse function,  $f^{-1}(x)$  1
- (iii) State the domain of  $f^{-1}(x)$  1
- (iv) Find the equation of  $f^{-1}(x)$  2
- (v) For what value(s) of  $x$  does  $f(x) = f^{-1}(x)$ ? 2
- (vi) Evaluate  $f^{-1}(f(-1))$  1
- b) Sketch the function  $y = \sin^{-1}x$  showing clearly the domain and range. 2

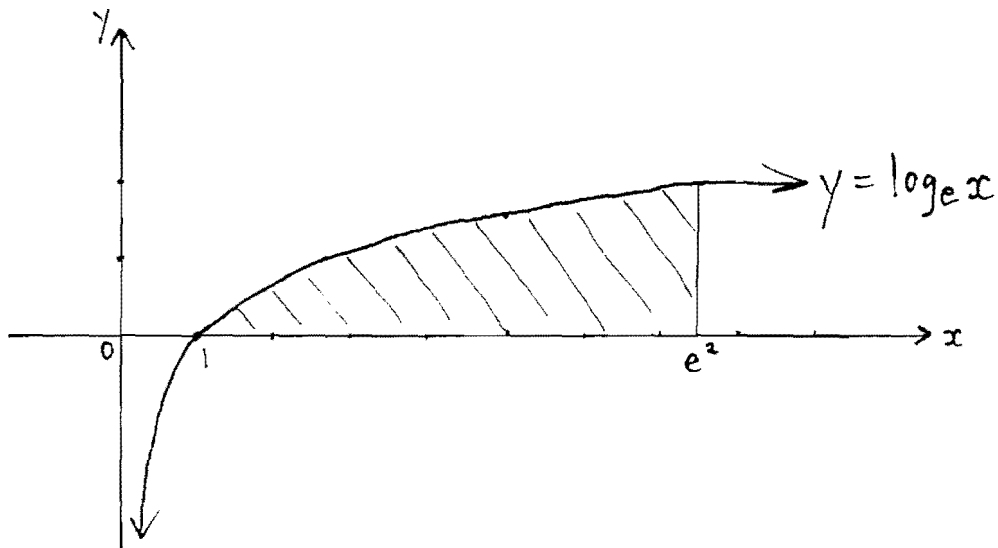
**Question 3****Marks**

Differentiate

- |       |                                                        |   |
|-------|--------------------------------------------------------|---|
| (i)   | $y = e^{3x} \log_e 3x$                                 | 2 |
| (ii)  | $y = \log_e \left( \frac{x-2}{x} \right)$              | 2 |
| (iii) | $y = \log_3 x$                                         | 2 |
| (iv)  | $y = \log_e (\cos x)$ (give answer in simplified form) | 2 |
| (v)   | $y = \tan (\log_e x)$                                  | 2 |

**Question 4**

- |    |                                                                                          |   |
|----|------------------------------------------------------------------------------------------|---|
| a) | (i) Express $\sin^2 x$ in terms of $\cos 2x$                                             | 1 |
|    | (ii) Use this result or otherwise to evaluate $\int_0^{\frac{\pi}{4}} \sin^2 2x dx$      | 3 |
| b) | Differentiate $\sin^2 3x$ and hence evaluate $\int_0^{\frac{\pi}{6}} \sin 3x \cos 3x dx$ | 3 |
| c) |                                                                                          | 3 |



Find the shaded area and leave it in exact form.

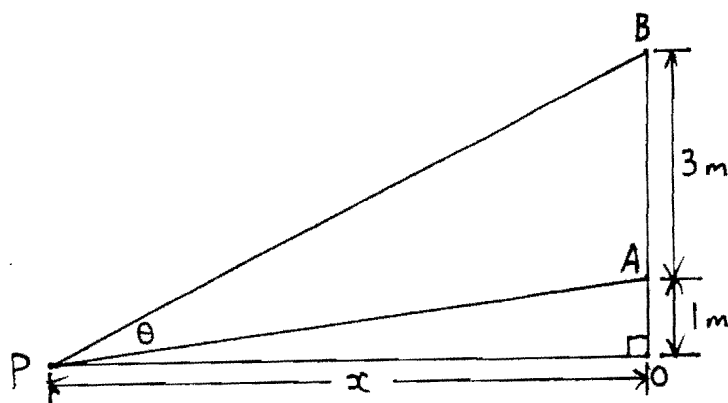
### Question 5

Marks

- a) Find the general solution for  $\sin\theta = \frac{1}{\sqrt{2}}$  2
- b) Differentiate (i)  $y = \tan^{-1}3x$  2
- (ii)  $y = \sin^{-1}(\cos x)$  2
- c) Sketch  $y = 1 - 3\sin 2x$  between  $0 \leq x \leq 2\pi$  3
- d) Sketch  $y = \cos^{-1}(\cos x)$  2

### Question 6

- a) A spherical balloon is being inflated so that the surface area is increasing at the rate of  $0.3\text{cm}^2\text{s}^{-1}$ . When the balloon's radius is 4cm, find the rate of increase.  $\left[v = \frac{4}{3}\pi r^3, SA = 4\pi r^2\right]$
- i) In the radius (correct to 3 decimal places) 2
- ii) In the volume (correct to one decimal place) 2
- b) In the diagram, a vertical pole AB, 3 metres high, is placed on top of a 1 metre high support. The pole subtends an angle of  $\theta$  radians at the point P, which is  $x$  metres from the base O of the support.



- (i) Show that  $\theta = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{1}{x}$  2
- (ii) Show that  $\theta$  has a stationary point when  $x = 2$ . 3
- Assume it is a maximum.
- (iii) Deduce that the maximum angle subtended at P is  $\theta = \tan^{-1} \frac{3}{4}$ . 2

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## 2009 Ext. 1 Task 3 Solutions

## Question 1

$$a) \log_e x + \log_e (x-3) = \log_e 4$$

$$\log_e x(x-3) = \log_e 4$$

$$x^2 - 3x = 4$$

$$x^2 - 3x - 4 = 0$$

$$(x+1)(x-4) = 0$$

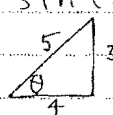
$$x = -1 \text{ or } 4$$

but  $x$  must be  $> 0$

$$\therefore x = 4$$

$$ii) \sin(2 \cos^{-1} \frac{4}{5})$$

Let  $\cos^{-1} \frac{4}{5} = \theta$



$$\therefore \sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \times \frac{3}{5} \times \frac{4}{5}$$

$$= \frac{24}{25}$$

$$iii) \int \frac{dx}{\sqrt{1-9x^2}}$$

$$\frac{1}{3} \int \frac{dx}{\sqrt{(\frac{1}{3})^2 - x^2}}$$

$$\frac{1}{3} \sin^{-1} \frac{x}{\frac{1}{3}}$$

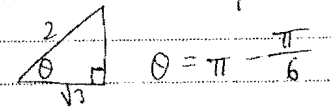
$$\frac{1}{3} \sin^{-1} 3x + c$$

$$b) i) \cos^{-1}(-\frac{\sqrt{3}}{2})$$

Range of  $\cos^{-1}$  is

$$0 \leq \theta \leq \pi$$

is in 2nd quadrant



$$\theta = \pi - \frac{\pi}{6}$$

$$\theta = \frac{5\pi}{6}$$

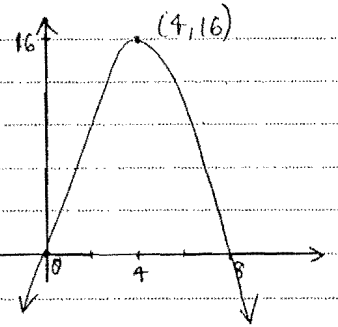
$$c) i) \int \frac{2}{9+x^2} dx$$

$$\frac{2}{3} \int \frac{1}{9+x^2} dx$$

$$= \frac{2}{3} \tan^{-1} \frac{x}{3} + c$$

## Question 2

$$a) i) f(x) = 8x - x^2$$



$$ii) x \geq 4$$

$$iv) y = 8x - x^2$$

$$x = 8y - y^2$$

$$y^2 - 8y + 16 = -x + 16$$

$$(y-4)^2 = -x + 16$$

$$y-4 = \pm \sqrt{16-x}$$

$$y = 4 \pm \sqrt{16-x}$$

$$iii) x \leq 16$$

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v) They intersect on

$$y = x$$

$$\therefore \text{Solve } y = x \text{ and } y = 8x - x^2$$

$$x = 8x - x^2$$

$$x^2 = 7x$$

$$x(x-7) = 0$$

$$x = 7 \Rightarrow (7, 7)$$

$$vi) f^{-1}(f(-1))$$

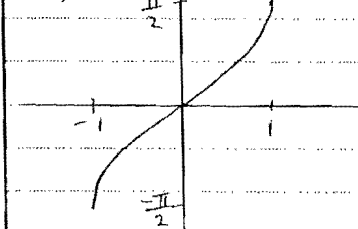
$$= f^{-1}(-9)$$

$$= 4 + \sqrt{16 - (-9)}$$

$$= 4 + 5$$

$$= 9$$

$$b) y = \sin^{-1} x$$



## Question 3

$$ci) y = e^{3x} \log_e 3x$$

$$y' = e^{3x} \times \frac{3}{3x} + 3e^{3x} \log_e 3x$$

$$= \frac{e^{3x}}{x} + 3e^{3x} \log_e 3x$$

$$cii) y = \log_e \left( \frac{x-2}{x} \right)$$

$$= \log_e (x-2) - \log_e x$$

$$y' = \frac{1}{x-2} - \frac{1}{x}$$

$$ciii) y = \log_3 x$$

$$y = \frac{\log_e x}{\log_e 3}$$

$$y' = \frac{1}{x \log_e 3}$$

$$civ) y = \log_e (\cos x)$$

$$y' = \frac{-\sin x}{\cos x}$$

$$y' = -\tan x$$

$$v) y = \tan(\log_e x)$$

$$y' = \sec^2(\log_e x) \times \frac{1}{x}$$

$$y' = \frac{\sec^2(\log_e x)}{x}$$

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## Question 4

$$a) (i) \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$(ii) \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 2x \, dx$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1 - \cos 4x}{2} \, dx$$

$$\frac{1}{2} \left[ x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{4}}$$

$$\frac{1}{2} \left[ \frac{\pi}{4} - 0 - (0) \right]$$

$$= \frac{\pi}{8}$$

$$b) \frac{d}{dx} (\sin^2 3x) = 2 \sin 3x \cos 3x \times 3$$

$$= 6 \sin 3x \cos 3x$$

$$\therefore \frac{1}{6} \frac{d}{dx} (\sin^2 3x) = \sin 3x \cos 3x$$

$$\therefore \frac{1}{6} \int_0^{\frac{\pi}{6}} \frac{d}{dx} (\sin^2 3x) \, dx$$

$$= \int_0^{\frac{\pi}{6}} \sin 3x \cos 3x \, dx$$

$$LHS = \frac{1}{6} \left[ \sin^2 3x \right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{6} \left( \sin^2 \frac{\pi}{2} \right)$$

$$= \frac{1}{6}$$

c) Area shaded = Rectangle - Area to y axis

$$= e^2 \times 2 - \int_0^2 x \, dy$$

$$= 2e^2 - \int_0^2 e^y \, dy$$

$$= 2e^2 - [e^y]_0^2$$

$$= 2e^2 - (e^2 - e^0)$$

$$= e^2 + 1$$

## Question 5

$$a) \sin \theta = \frac{1}{\sqrt{2}}$$

$$\theta = n\pi + (-1)^n \sin^{-1} \frac{1}{\sqrt{2}}$$

$$\theta = n\pi + (-1)^n \times \frac{\pi}{4}$$

$$b) (i) y = \tan^{-1} 3x$$

$$y = \tan^{-1} \frac{x}{3}$$

$$y' = \frac{\frac{1}{3}}{x^2 + (\frac{1}{3})^2}$$

$$= \frac{3}{9x^2 + 1}$$

$$(ii) y = \sin^{-1}(\cos x)$$

$$y' = \frac{-\sin x}{\sqrt{1 - \cos^2 x}}$$

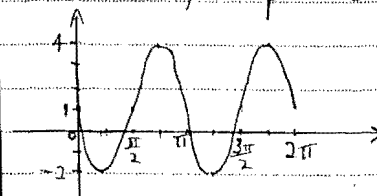
$$y' = -1$$

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$$c) y = 1 - 3 \sin 2x$$

Period  $\pi$ , Amplitude 3

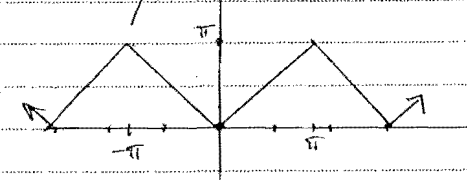


$$d) y = \cos^{-1}(\cos x)$$

Domain: All real  $x$

Range:  $0 \leq y \leq \pi$

Since  $\cos x$  is even, so is  $y$ .



## Question 6

$$a) \frac{dA}{dt} = 0.3 \quad A = 4\pi r^2 \quad (ii) \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$0.3 = 8\pi r \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{0.3}{8\pi \times 4}$$

$$\frac{dr}{dt} = 0.003 \text{ cm/s}$$

$$= 4\pi r^2 \times 0.003$$

$$= 4\pi \times 4^2 \times 0.003$$

$$= 0.6 \text{ cm}^3/\text{s}$$

$$b) (i) \tan \angle BPO = \frac{4}{x} \Rightarrow \angle BPO = \tan^{-1} \frac{4}{x} \quad \theta = \angle BPO - \angle APO$$

$$\tan \angle APO = \frac{1}{x} \Rightarrow \angle APO = \tan^{-1} \frac{1}{x} \therefore \theta = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{1}{x}$$

$$(ii) \frac{d\theta}{dx} = \frac{-\frac{4}{x^2}}{1 + (\frac{4}{x})^2} - \frac{-\frac{1}{x^2}}{1 + (\frac{1}{x})^2}$$

$$= \frac{-4}{x^2 + 16} + \frac{1}{x^2 + 1}$$

$$= \frac{-4x^2 - 4 + x^2 + 16}{(x^2 + 16)(x^2 + 1)}$$

$$\frac{d\theta}{dx} = 0 \text{ if } -3x^2 + 12 = 0$$

$$x^2 = 4$$

$$x = 2, (x > 0)$$

$$(iii) \text{Sub. } x = 2 \text{ into}$$

$$\theta = \tan^{-1} \frac{4}{2} - \tan^{-1} \frac{1}{2}$$

$$\tan \theta = \tan [\tan^{-1} 2 - \tan^{-1} \frac{1}{2}]$$

$$= \frac{\tan(\tan^{-1} 2) - \tan(\tan^{-1} \frac{1}{2})}{1 + \tan(\tan^{-1} 2) \tan(\tan^{-1} \frac{1}{2})}$$

$$= \frac{2 - \frac{1}{2}}{1 + 2 \times \frac{1}{2}}$$

$$= \frac{1\frac{1}{2}}{2}$$

$$= \frac{1\frac{1}{2}}{2}$$

$$\therefore \theta = \tan^{-1} \frac{3}{4}$$